A WEB of Sharing Trust–Responsibility Toward a Trustless Future vol.2

AI Governance Paper with Architecture-Specific Extension
Second Edition — Version B

Document Control Information

Document Identifier: APOD-TR-003
Title: *A WEB of Sharing Trust–Responsibility Toward a Trustless Future Vol.2*
Edition: Second Edition — Version B
Date of Issue: 11 January 2026
Status: Public Review Draft (Standards Exploration)
Author: Anthony Siljeg
Affiliation: Alnotrea Labs / Policy of Developments - Access PoD *Initiative.*

Distribution Statement

This publication is released as a **Public Review Draft (Standards Exploration)**.

It may be used for:

- academic research and comparative analysis,
- policy development and regulatory consultation,
- standards exploration and working-draft discussion, and
- evaluation of approaches to AI governance that move beyond static, documentation-based compliance toward **engineered, runtime trust mechanisms**.
- Feedback is invited to support refinement prior to subsequent editions.

This document does **not** constitute regulatory guidance, certification, or legal advice. It is intended to support informed review, discussion, and refinement of governance architectures, assurance models, and scoring methodologies for AI systems.

Second Edition Note — Version B

(Architecture-Specific Extension)

This Second Edition — Version B builds upon **Volume 2, Version A**, which defines a **model-agnostic, execution-independent AI governance framework** based on continuous, evidence-derived assurance.

Version B introduces an **optional, architecture-specific extension**, using the Q-Pathformer system solely as a **reference implementation**. Its purpose is to demonstrate how the Version A governance framework can be instantiated at execution depth within a concrete AI system, **without redefining governance principles, assurance semantics, or Compliance Star Certificate (CSC) logic**.

Version B is a working draft.
Readers are strongly encouraged to review Version A first, which remains the authoritative reference for governance authority, terminology, and constraints.

The stable Version A Public Review Draft is available at:
https://policyofdevelopments.org/whitepaper/vol2-a-web-of-sharing-trust-responsibility-toward-a-trustless-future

Version B extends Version A by:

- illustrating runtime governance attachment points,
- exploring execution-level telemetry and provenance,
- and stress-testing governance–execution separation under realistic system conditions.

Accordingly, Version B should be read as an **exploratory, thinking-forward extension**, not as a finalized specification, certification scheme, or standards proposal.

Version B is shared to support:

- early technical and architectural review,
- standards and policy discussion, and
- informed feedback prior to consolidation.

Where differences or ambiguities arise, **Version A prevails** until a subsequent consolidated edition (Version C or later) is formally issued.

Key Characteristics of Version B

- **Subordinate to Version A**
  All governance authority, assurance semantics, and CSC scoring logic remain defined exclusively by Version A.
- **Architecture as Execution Substrate, Not Authority**
  Q-Pathformer is used only to illustrate execution boundaries, telemetry generation, and lifecycle controls. Governance authority remains external and is exercised through the Access Policy-on-Demand (PoD) framework.
- **Strict Scope Discipline**
  Version B explicitly excludes cryptographic realization, validation internals, training algorithms, onboarding procedures, and performance benchmarking. These exclusions are intentional to preserve security and standards-exploration integrity.
- **Governance–Execution Separation**
  Version B formalizes a clear separation between governance authorization (Access PoD), controlled execution (architecture-specific APIs), and assurance interpretation (CSC), as a core safety and integrity requirement.

Relationship to Subsequent Editions

Version B is **not** an onboarding guide, operational manual, or implementation specification.

A subsequent **Version C** is planned to address Q-Pathformer-specific onboarding, operational workflows, and deployment considerations. That material will remain subordinate to the governance framework defined in Version A and instantiated in Version B.

Intended Use

Version B is intended for:

- standards exploration and review,
- policy and regulatory consultation,
- architectural feasibility assessment, and
- informed critique of runtime AI governance models.

It does **not** constitute regulatory guidance, certification, or legal advice.

Tables of Content

Version B — Architecture-Specific Extension (Reference Implementation)

Second Edition Note — Version B……2

Executive Summary……6

1. **Introduction**
   1.1 Purpose of the Architecture-Specific Extension……9
   1.2 Relationship to Version A Framework……10
2. **Architecture Context and Scope**
   2.1 Q-Pathformer as a Reference Implementation……12
   2.2 Multi-State Learning Model (Informative)……13
   2.3 Scope Boundaries and Non-Goals……13
3. **Governance Integration with Access PoD**
   3.1 Policy-as-Code Alignment……15
   3.2 Simulation AI-Powered Policies (SAPP) Integration……15
   3.3 Telemetry and Evidence Continuity……16
   3.4 Delegation and Execution Boundaries……17
   3.5 Authority Preservation and Non-Substitution……17
4. **UGCCMT Provenance Model**
   4.1 Conceptual Overview of UGCCMT……19
   4.2 Modality Coverage and Majority Modalities……20
   4.3 Lineage, Traceability, and Audit Semantics……20
   (Cryptographic realization referenced but restricted)
5. **API Governance and Controlled Execution Interfaces**
   5.1 Governance–Execution Separation……22
   5.2 APIs as Policy-Bound Nodes……22
   5.3 Data Procurement and Training Proposals……23
   5.4 Controlled Execution via Q-Pathformer API……23
   5.5 Third-Party Model APIs and Standards Participation……24
   5.6 Catastrophic Overwrite Prevention……24
   5.7 Relationship to Restricted Specifications……25
6. **Compliance Star Certificate (CSC) Integration**
   6.1 CSC Evidence Inputs from Architecture Telemetry……26
   6.2 Assurance Boundaries and Interpretation……26
   6.3 Revocation and Degradation Semantics……27
   6.4 Evidence Packaging and Proportional Disclosure (Informative)……27

7. **Security, Trust, and Responsibility Boundaries**
   7.1 What Is Governed in Version B……29
   7.2 What Is Explicitly Out of Scope……..29
   7.3 Responsibility Allocation and Non-Transferability……30
   7.4 Relationship to Restricted Specifications……..30
   7.5 Relationship to Version C and Operational Documentation……31
8. **Forward-Looking Considerations (Non-Normative)**
   8.1 Interoperability and Ecosystem Evolution……32
   8.2 Future Cryptographic and Privacy Enhancements……32
   8.3 Transition Considerations for Version C……33
   8.4 Version D — Governance Stress-Testing and Ethical Evaluation……33

9. Conclusion
   9.1 Engineering Trust Before Impact……35
   9.2 Standards, Conformance, and Ongoing Evolution……35

Conformance and Interoperability Clause……37
Appendix A — Methods (Normative)……38 - 43
Appendix B — Informative References and Supporting Artifacts……44 - 45

Clarifying Inserts (Informative — Non-Normative)……46 - 47

How This Framework Differs from Certification Schemes……49
Cover Note — Regulator Briefing (Informative)……50 - 52
Regulator Briefing — Runtime AI Governance and Continuous Assurance……53 - 56

Executive Summary

As artificial intelligence systems become adaptive, compositional, and continuously updated, governance approaches based on static documentation, point-in-time audits, and declarative compliance are no longer sufficient to sustain trust.

Volume 2, Version A introduced a model-agnostic framework for **engineering trust as a runtime system property**, rather than treating it as an administrative outcome. That framework formalized three complementary mechanisms:

1. **Simulation AI-Powered Policies (SAPP)** for pre-deployment and early-deployment risk discovery,
2. a **Policy-on-Demand (PoD) runtime governance engine** for enforceable control execution, and
3. a telemetry-derived **Compliance Star Certificate (CSC)** for continuous, revocable assurance.

CSC assurance is derived exclusively from observed system behavior, not documentation, intent, or attestation.

Purpose of Version B

This Version B document introduces an **architecture-specific extension** using the Q-Pathformer system as a reference implementation. Its purpose is to demonstrate that the Version A governance framework can be **operationalized at execution depth** without embedding governance authority into system architecture.

Version B shows:

- how runtime governance controls can attach to real execution boundaries,
- how governance-relevant telemetry can be generated continuously,
- how provenance and lineage can be preserved across adaptive lifecycles, and
- how assurance can degrade or revoke when evidence no longer supports trust claims.

The inclusion of Q-Pathformer does not modify, constrain, or replace the architecture-agnostic applicability of Version A.

Core Contributions

This document makes four primary contributions to AI governance practice:

1. **Governance–Execution Separation**
   Governance authority (Access PoD) is explicitly separated from execution capability (architecture-specific APIs). No system component may propose, approve, and execute changes unilaterally.
2. **Provenance as a Governance Substrate**
   The UGCCMT (Unit–Group–Colour–Code–Measure–Train) model is introduced as a structured provenance and lineage abstraction, enabling traceability, attribution, and audit across heterogeneous data modalities and lifecycle phases.
3. **API-Level Governance as a Security Control**
   Execution APIs are treated as policy-bound nodes rather than neutral interfaces. This design prevents uncontrolled data ingestion, limits blast radius, and protects against catastrophic overwrite in long-lived AI systems.
4. **Continuous, Revocable Assurance**
   CSC integration demonstrates how assurance can be computed continuously from telemetry, interpreted within declared scope, and revoked when evidence degrades—without reliance on static certification.
5. In Version B, CSC assurance is treated not only as a computed signal, but as a **portable governance evidence artifact**. CSC inputs may be grouped into governed evidence bundles of varying sensitivity, enabling proportional disclosure, time-bound access, and third-party confirmation without exposing raw telemetry or protected operational detail. This evidence-bundling approach is illustrative of how continuous assurance may be operationalized in regulated environments while preserving confidentiality, minimization, and audit integrity.

Scope, Limitations, and Onboarding Status

Version B is intentionally scoped.

It does **not** specify:

- cryptographic mechanisms or security parameters,
- training or optimization methods,
- operational onboarding procedures, or
- regulatory certification criteria.

These elements are either governed by restricted specifications or reserved for subsequent documentation.

Intended Audience

This document is intended for:

- policymakers and regulators evaluating runtime AI governance models,
- standards bodies exploring evidence-based assurance mechanisms,
- researchers assessing governance feasibility in adaptive systems, and
- system designers seeking to separate trust enforcement from model architecture.

Q-Pathformer Onboarding Status and Infrastructure Readiness (Informative)

For Version B, the Q-Pathformer system is used solely as a **reference execution substrate** to validate governance attachment points, execution boundaries, and telemetry generation, rather than as a production-ready model deployment. Evaluation and onboarding are intentionally conducted in **constrained environments**, including **laptop-class hardware and edge-equivalent execution contexts**, to prioritize correctness, observability, and reproducibility over scale or performance. This reflects a design goal of enabling Access PoD–governed execution and validation on modest hardware, supporting early standards review, independent testing, and distributed governance validation. Large-scale training, specialized hardware acceleration, and operational onboarding are out of scope for Version B and do not affect the validity of the governance framework or assurance semantics. Broader onboarding of Q-Pathformer is contingent on further validation and infrastructure testing, with an **indicative timeframe of mid to late 2026**, subject to successful outcomes.

Key Message

Trust in AI systems cannot be established through declaration, documentation, or one-time assessment.

It shall be engineered prior to impact, continuously observed during operation, and revoked when evidence no longer supports confidence.

Version B demonstrates that this approach is feasible in practice without embedding governance authority into AI architectures, and without constraining interoperability, accountability, or innovation.

1. Introduction

Artificial intelligence systems are increasingly deployed as **long-lived, adaptive infrastructures**, rather than static software artifacts. As these systems evolve through retraining, fine-tuning, data augmentation, and integration with external services, governance approaches based on static documentation, point-in-time audits, or declarative compliance statements become insufficient to ensure sustained trust.

Volume 2, Version A established a **model-agnostic governance framework** for AI systems based on three core mechanisms:

- **Simulation AI-Powered Policies (SAPP)** for pre-deployment and early-deployment risk discovery,
- a **Policy-on-Demand (PoD) runtime governance engine** for enforceable control execution, and
- a telemetry-derived **Compliance Star Certificate (CSC)** for continuous, revocable assurance.

That framework intentionally abstracted away from any specific AI architecture to ensure broad applicability across machine learning pipelines, foundation models, and agentic systems.

This Version B document introduces an **optional, architecture-specific extension**, using the Q-Pathformer system as a reference implementation. Its purpose is to demonstrate how the Version A governance framework may be instantiated at execution depth **without altering, constraining, or redefining the core governance model**. Within this framework, system onboarding is treated as a governed entry condition—establishing eligibility for controlled testing and evidence generation, rather than conferring trust, approval, or deployment readiness.

1.1 Purpose of the Architecture-Specific Extension

The purpose of this architecture-specific extension is **illustrative and confirmatory**, not normative with respect to AI system design.

Specifically, Version B aims to:

- demonstrate how runtime governance controls defined in Version A can be executed against a concrete AI architecture,
- show how governance telemetry may be generated from real execution pathways rather than inferred behavior,
- validate that the CSC assurance model remains applicable when governance is applied at execution boundaries, and

- clarify how provenance, lineage, and lifecycle controls can be preserved in adaptive training systems.

This extension **does not prescribe** how AI systems must be built. It does not establish performance requirements, architectural benchmarks, or preferred learning paradigms. Instead, it provides a **reference instantiation** that enables reviewers, policymakers, and engineers to reason concretely about governance behavior under realistic system conditions.

The inclusion of Q-Pathformer in Version B **shall not be interpreted** as an endorsement, certification, or requirement for adoption. The architecture is used solely to demonstrate feasibility and boundary conditions for the governance framework defined in Version A.

1.2 Relationship to Version A Framework

Version B is strictly **subordinate to and dependent upon** the Version A governance framework.

Accordingly:

- All governance principles, assurance definitions, and CSC semantics defined in Version A remain authoritative.
- Version B introduces no new governance domains, scoring factors, or certification criteria.
- Any architecture-specific telemetry described in this document maps directly to Version A operational factors and CSC evidence requirements.

Where Version A defines *what* governance outcomes must be achieved (e.g., continuous assurance, revocability, human-centric oversight), Version B demonstrates *how* those outcomes may be supported within a specific execution model.

This document therefore preserves a clear separation between:

- **governance intent** (Version A),
- **governance execution** (Version B), and
- **security or cryptographic implementation details**, which are intentionally excluded from public disclosure and governed by restricted specifications.

Nothing in Version B modifies the architecture-agnostic applicability of the Version A framework. Systems that do not resemble Q-Pathformer remain fully compatible with the governance model defined in Volume 2.

Normative Summary (Chapter 1)

- Version B is an optional, architecture-specific extension to the Version A governance framework.
- The Q-Pathformer system is used as a reference implementation only.
- Governance authority, assurance semantics, and CSC scoring remain defined by Version A.
- Architecture-specific content in Version B shall not be interpreted as prescriptive, mandatory, or exclusive.
- Security and cryptographic mechanisms are intentionally out of scope for this public document.

---

2. Architecture Context and Scope

This chapter defines the architectural context in which the Version A governance framework is instantiated for the purposes of this extension. It clarifies the role of the Q-Pathformer system as a reference implementation, establishes the abstraction level at which architecture is discussed, and explicitly delineates what is within and outside the scope of Version B.

The intent of this chapter is to ensure that architecture is treated as an **execution substrate for governance**, rather than as a source of governance authority, certification, or policy definition.

2.1 Q-Pathformer as a Reference Implementation

For Version B, the Q-Pathformer system is used as a **reference implementation** to illustrate how runtime governance, telemetry generation, and assurance mechanisms defined in Version A may be applied within a concrete AI training and execution environment.

Q-Pathformer is selected for this purpose because it exhibits characteristics common to contemporary and emerging AI systems, including:

- multi-phase training and retraining workflows,
- adaptive data ingestion and curation,
- modular execution components,
- integration of multiple data modalities, and
- long-lived model lifecycles with iterative updates.

These characteristics make it suitable for demonstrating governance behavior under realistic conditions without implying that such characteristics are unique or required.

Importantly, the use of Q-Pathformer in this document **does not establish it as a governance framework**, nor does it elevate it above other AI architectures. Governance authority remains fully vested in the Version A framework (SAPP, Access PoD, CSC), which is designed to operate independently of any specific model architecture.

Informative Note — Subsequent Onboarding Documentation
A separate Volume 2 / Version C document is planned to address **Q-Pathformer onboarding, operational mechanisms, and implementation detail**. That material is intentionally excluded from Version B to preserve the standards-exploration scope of this document and specific security mechanisms.

2.2 Multi-State Learning Model (Informative)

Q-Pathformer employs a multi-state learning model to organize training, refinement, and execution activities across distinct lifecycle phases. In this document, the multi-state model is described **for contextual understanding only**.

At a high level, the architecture differentiates between:

- pre-training and abstraction phases,
- grouping and consolidation phases, and
- execution and fine-tuning phases.

These states provide natural enforcement points for governance controls, telemetry capture, and lifecycle assurance. However, **no assumptions are made** in this document regarding:

- internal learning algorithms,
- optimization strategies,
- parameter update mechanisms, or
- performance characteristics.

The multi-state description is included solely to explain **where governance controls attach**, not **how learning is performed**.

No governance requirement in Version A or Version B depends on the existence of a specific state model. Equivalent attachment points may exist in other architectures with different internal structures.

2.3 Scope Boundaries and Non-Goals

To avoid ambiguity, this section explicitly defines the scope boundaries of Version B.

2.3.1 In Scope

Version B addresses architecture only to the extent necessary to:

- identify execution and control boundaries,
- demonstrate telemetry generation aligned with Version A operational factors,
- show how provenance and lineage signals may be preserved across lifecycle events, and
- illustrate how governance decisions can be enforced without modifying learning behavior.

Architecture is discussed at the level of **interfaces, boundaries, and observable behavior**, not internal mechanics.

2.3.2 Out of Scope

The following are explicitly out of scope for Version B:

- training algorithms, loss functions, or optimization logic,
- model parameter structures or weight update strategies,
- cryptographic mechanisms, key management, or validation internals,
- security thresholds, re-derivation logic, or incident handling playbooks,
- performance benchmarking or comparative evaluation, and
- operational onboarding or deployment procedures.

These elements are either governed by restricted specifications or reserved for subsequent documentation.

2.3.3 Relationship to Version C (Onboarding and Mechanisms)

Version B shall not be treated as an onboarding guide, operational manual, or implementation specification for Q-Pathformer.

A subsequent **Version C / Volume 2 document** is intended to address onboarding, operational workflows, and architecture-specific mechanisms in detail. That document will be explicitly scoped to Q-Pathformer and will not modify the governance framework defined in Version A or instantiated in Version B.

The separation between Version B and Version C is intentional and necessary to ensure that standards exploration, governance validation, and operational enablement remain distinct and independently reviewable.

Normative Summary (Chapter 2)

- Q-Pathformer is used as a reference implementation only.
- Governance authority remains defined by Version A.
- Architecture is discussed at the level of execution boundaries and telemetry attachment points.
- Learning mechanics, cryptography, and onboarding are out of scope.
- Q-Pathformer onboarding and operational detail are reserved for Version C / Volume 2.

3. Governance Integration with Access PoD

This chapter defines how the Version A governance framework—comprising Simulation AI-Powered Policies (SAPP), the Access Policy-on-Demand (PoD) engine, and the Compliance Star Certificate (CSC)—is integrated with the reference architecture for the purposes of this extension.

The objective of this chapter is to demonstrate how governance controls are **executed, observed, and evidenced at runtime**, while preserving the separation between governance authority and system implementation established in earlier sections.

3.1 Policy-as-Code Alignment

Within the Version A framework, governance policies are defined as **executable policy-as-code artifacts**. These policies encode constraints, thresholds, and escalation conditions that are enforceable during system operation rather than relying on post-hoc review or documentation.

In the context of this reference implementation:

- Policy-as-code artifacts are evaluated and enforced by the Access PoD engine.
- The reference architecture exposes explicit execution boundaries at which policies may be applied.
- Policy evaluation outcomes are emitted as machine-readable telemetry for assurance purposes.

Policy-as-code alignment does **not** require the architecture to implement governance logic internally. Instead, the architecture must expose sufficient interfaces and state signals to allow external policy execution to be effective.

Governance policies shall therefore be **architecture-aware but architecture-independent**: aware of execution boundaries, yet not embedded within learning or inference logic.

3.2 Simulation AI-Powered Policies (SAPP) Integration

Simulation AI-Powered Policies (SAPP) provide the pre-deployment and early-deployment governance layer of the framework. Their role within the reference architecture is to enable **risk discovery and control validation before and during live execution**.

For the purposes of this extension:

- SAPP evaluates proposed training events, dataset changes, and configuration updates under simulated conditions.
- Simulation outputs generate telemetry consistent with Version A evidence requirements.
- Human review remains mandatory for elevated-risk simulation outcomes.

Integration with the reference architecture occurs at **proposal and transition points**, not within learning execution itself. SAPP does not modify model behavior; it evaluates governance assumptions prior to authorizing execution through Access PoD.

This ensures that novelty, scale, or apparent performance improvements do not bypass governance validation.

3.3 Telemetry and Evidence Continuity

A core requirement of Version A is that governance assurance be derived from **continuous, observable evidence**. In this extension, the reference architecture supports this requirement by enabling telemetry continuity across lifecycle events.

Specifically:

- Each governance-relevant action (e.g., data access, training execution, delegation to external models) produces structured telemetry.
- Telemetry is attributable to defined enforcement points and policy decisions.
- Telemetry persists across system updates, retraining events, and integration changes.

The architecture is not required to compute CSC scores or interpret assurance outcomes. Its responsibility is limited to **producing accurate, attributable telemetry** that can be consumed by Access PoD and downstream assurance mechanisms.

Where telemetry is incomplete, delayed, or unavailable, CSC scoring shall be constrained or suspended in accordance with Version A requirements. In practice, this continuity presumes deterministic artifact handling—where governance evidence is generated only after controlled fetch, unpacking, and verification into an inspectable execution workspace. Telemetry continuity further enables governance systems to assemble **contextual evidence packages** derived from multiple CSC-relevant signals. Such packages may be formed to reflect different governance or regulatory sensitivity tiers, allowing oversight bodies to confirm the existence, integrity, and scope of operational evidence without requiring unrestricted access to underlying execution logs or model artifacts.

3.4 Delegation and Execution Boundaries

Modern AI systems frequently rely on delegated execution across internal components and external services. Governance integration therefore requires explicit control over **delegation boundaries**.

Within this framework:

- Access PoD governs whether execution may be delegated to internal or external components.
- Delegated execution shall not transfer governance responsibility.
- Telemetry from delegated actions shall be captured and attributed to the originating governance context.

The reference architecture supports this model by exposing delegation events as first-class governance signals. This enables governance continuity even when execution spans heterogeneous models, APIs, or service providers.

3.5 Authority Preservation and Non-Substitution

The integration described in this chapter shall not be interpreted as transferring governance authority to the reference architecture.

Accordingly:

- The architecture does not define policy, risk thresholds, or assurance criteria.
- It does not approve or reject governance outcomes independently.
- It does not issue or interpret CSC scores.

All such authority remains vested in the Version A governance framework.

This preservation of authority ensures that governance remains **portable, comparable, and interoperable** across architectures, rather than becoming embedded within any single implementation.

Normative Summary (Chapter 3)

- Governance policies shall be executable as policy-as-code through Access PoD.

- SAPP shall evaluate risk prior to and during execution transitions.
- The architecture shall expose governance-relevant execution boundaries.
- Telemetry shall be continuous, attributable, and machine-readable.
- Delegation shall not obscure accountability.
- Governance authority shall not be transferred to the architecture

---

4. UGCCMT Provenance Model

This chapter introduces the **UGCCMT provenance model** as the architecture-specific mechanism used in this reference implementation to generate structured, machine-readable lineage and accountability signals for governance and assurance.

UGCCMT—short for **Unit–Group–Colour–Code–Measure–Train**—is not a governance framework, scoring system, or certification mechanism. It is a **provenance abstraction** designed to ensure that governance-relevant activity within an AI system can be **observed, attributed, and audited** across the system lifecycle.

Within Version B, UGCCMT is presented **at the semantic and structural level only**. Cryptographic sealing, validation, and enforcement mechanisms associated with UGCCMT are intentionally excluded from this public document and governed by restricted specifications.

4.1 Conceptual Overview of UGCCMT

UGCCMT provides a structured method for describing how data, transformations, and learning events propagate through an AI system over time.

At a conceptual level, UGCCMT defines six orthogonal dimensions:

- **Unit** – the smallest addressable element subject to governance (e.g., a data element, token span, or interaction unit),
- **Group** – a logical aggregation of units for lifecycle or governance purposes,
- **Colour** – a categorical context or risk lens applied to the unit or group,
- **Code** – the rule, constraint, or classification context under which processing occurs,
- **Measure** – quantitative or qualitative signals generated through processing, and
- **Train** – the lifecycle phase or learning context in which the unit participates.

Together, these dimensions enable governance systems to answer fundamental questions such as:

- *What was processed?*
- *Under what constraints?*
- *In which lifecycle phase?*
- *With what observed outcomes?*

UGCCMT does not dictate how learning occurs. It ensures that **whatever occurs can be traced, contextualized, and reviewed**.

4.2 Modality Coverage and Majority Modalities

UGCCMT is modality-agnostic. It applies consistently across heterogeneous data types and execution contexts.

For the purposes of this reference implementation, UGCCMT is applied across a set of **majority modalities** that collectively represent the dominant share of training, evaluation, and execution activity in contemporary AI systems. These include, but are not limited to:

- natural language text,
- source code and structured markup,
- static images,
- audio and speech data,
- video and sequential visual data,
- tabular and structured datasets,
- sensor and time-series data,
- structured documents and forms, and
- geospatial or location-based data.

Each modality is treated uniformly at the provenance level. Differences in data type or representation do not alter the UGCCMT dimensions; they influence only how units and measures are instantiated.

Hybrid or composite modalities may generate linked UGCCMT records, preserving lineage across multimodal transformations without collapsing provenance into a single representation.

4.3 Lineage, Traceability, and Audit Semantics

UGCCMT is designed to support **continuous lineage**, rather than point-in-time traceability.

Lineage within this model means that governance systems can reconstruct:

- how a unit entered the system,
- which transformations or training phases it participated in,
- which policy constraints were applicable at each stage, and
- which measurable outcomes were observed.

UGCCMT records are generated as an **inherent output of execution**, not as an after-the-fact annotation process. This ensures that provenance reflects actual system behavior rather than reconstructed intent.

UGCCMT does not itself perform audits, assign trust scores, or issue assurance claims. Instead, it provides the **semantic substrate** upon which:

- Access PoD enforces runtime governance,
- SAPP evaluates risk through simulation, and
- CSC derives evidence-based assurance signals.

Cryptographic sealing, validation, and tamper-resistance mechanisms that protect UGCCMT records are specified separately in restricted documentation and are intentionally out of scope for Version B.

Normative Summary (Chapter 4)

- UGCCMT defines a provenance and lineage abstraction, not a governance framework.
- The model is modality-agnostic and lifecycle-aware.
- UGCCMT records enable attribution, traceability, and audit across execution phases.
- UGCCMT does not prescribe learning behavior or performance characteristics.
- Cryptographic realization and enforcement of UGCCMT records are out of scope for this document.

---

5. API Governance and Controlled Execution Interfaces

This chapter defines the governance principles, control boundaries, and execution constraints applicable to application programming interfaces (APIs) used for data procurement, training execution, and model lifecycle operations within the reference implementation.

APIs are treated within this framework not as neutral transport mechanisms, but as **governance-critical execution boundaries**. Their design, scope, and authorization model directly affect the integrity, traceability, and safety of governed AI systems.

5.1 Governance–Execution Separation

A fundamental requirement of this framework is the **explicit separation of governance authority from execution capability**.

Accordingly:

- **Access PoD** shall serve as the governance, proposal, evaluation, and authorization layer.
- **Execution APIs**, including the Q-Pathformer API, shall operate solely as controlled execution interfaces.

No API endpoint shall be permitted to simultaneously:

1. propose new datasets, training objectives, or configuration changes;
2. approve such proposals; and
3. execute training, ingestion, or artifact modification.

This separation is a mandatory safeguard against uncontrolled system evolution and prevents governance bypass through API misuse or misconfiguration.

5.2 APIs as Policy-Bound Execution Nodes

Each governed API interface shall be treated as a **policy-bound execution node**.

An execution node is defined as an interface that:

- receives requests only through authenticated and authorized governance pathways,
- enforces declared scope, purpose, and lifecycle constraints prior to execution, and
- emits machine-readable telemetry as a first-class output.

Execution nodes shall not operate as general-purpose ingress points. Instead, they shall be explicitly scoped to:

- approved datasets and dataset versions,
- approved training or execution objectives,
- approved lifecycle stages (e.g., evaluation, training, promotion), and
- approved assurance contexts.

This node-based design limits the blast radius of execution-level failures and supports fine-grained attribution and audit.

5.3 Data Procurement and Training Proposals

All proposals for:

- new data procurement,
- dataset updates or extensions,
- retraining, fine-tuning, or harmonization events, and
- material configuration or lifecycle changes,

shall originate within the **Access PoD governance workflow**.

Such proposals shall:

- be evaluated against applicable policy constraints and sector profiles,
- be subjected to Simulation AI-Powered Policies (SAPP) where required, and
- require explicit authorization prior to execution.

Execution APIs, including the Q-Pathformer API, **shall not independently source, approve, or prioritize training data**.

This constraint ensures that data governance remains upstream of execution and that provenance integrity is preserved across the system lifecycle.

5.4 Controlled Execution via Q-Pathformer API

For the purposes of this architecture-specific reference implementation, the Q-Pathformer API serves as the **exclusive controlled execution interface** for approved training and dataset ingestion actions.

This exclusivity:

- does not confer governance authority,
- does not restrict standards participation, and
- does not preclude interoperability with third-party systems.

Rather, it establishes a **single, auditable execution surface** that enables:

- consistent enforcement of governance decisions,
- preservation of UGCCMT lineage and provenance continuity, and
- prevention of uncontrolled or accidental artifact overwriting.

Execution APIs shall act only on inputs that have been explicitly authorized through Access PoD and shall reject or defer all other requests.

5.5 Third-Party Model APIs and Standards Participation

Third-party model APIs may participate in the Access PoD ecosystem **subject to standards-mediated confirmation**.

Participation conditions shall include:

- explicit declaration of execution scope and intended use,
- compatibility with Access PoD governance signals and telemetry requirements, and
- confirmation by relevant standards bodies, sector authorities, or governance councils where applicable.

Third-party APIs shall not bypass Access PoD controls. Where external models are invoked, governance responsibility and assurance obligations shall remain with the integrating organization.

This model enables interoperability while preserving accountability and preventing ungoverned execution paths.

5.6 Catastrophic Overwrite Prevention

Within this framework, **catastrophic overwrite**—defined as uncontrolled, irreversible modification of governed datasets, model artifacts, or lineage records—is treated as a systemic security risk.

Accordingly:

- execution APIs shall enforce controlled versioning or immutability where applicable,
- overwriting of governed artifacts shall require explicit governance authorization, and
- rollback, revocation, and lineage continuity shall be supported as protective controls.

Catastrophic overwrite prevention is a **governance security objective**, not a performance optimization or implementation detail. It is essential for maintaining trust, auditability, and long-term system integrity. This constraint applies equally during testing, evaluation, and governance stress scenarios.

5.7 Relationship to Restricted Specifications

This chapter specifies governance roles, execution boundaries, and control objectives only.

Cryptographic enforcement mechanisms, validation workflows, key management logic, and low-level security controls that support API governance are defined in **restricted specifications** and are intentionally excluded from this public standards-exploration document.

Normative Summary (Chapter 5)

- Governance authority and execution capability shall be separated.
- APIs shall function as policy-bound execution nodes.
- Data procurement and training proposals shall originate in Access PoD.
- Execution APIs shall act only on approved scopes.
- Third-party APIs may participate subject to standards-mediated confirmation.
- Catastrophic overwrite prevention shall be treated as a core security objective.
- Cryptographic realization is out of scope for this document.

---

6. Compliance Star Certificate (CSC) Integration

This chapter defines how the **Compliance Star Certificate (CSC)** is integrated with the reference architecture to provide **continuous, telemetry-derived assurance**. CSC integration in Version B demonstrates how execution-level telemetry and provenance signals are transformed into interpretable trust indicators, while preserving the authority, semantics, and scoring logic defined in Version A.

CSC is not a static certification, endorsement, or approval. It is a **dynamic assurance signal** derived exclusively from observed system behavior.

6.1 CSC Evidence Inputs from Architecture Telemetry

CSC evidence shall be derived exclusively from **telemetry generated during simulation and runtime operation**, in accordance with Version A requirements.

Within the reference implementation, CSC evidence inputs include:

- governance enforcement telemetry emitted by Access PoD at defined execution boundaries,
- SAPP simulation outputs associated with proposed changes or elevated-risk scenarios,
- UGCCMT provenance and lineage records describing lifecycle participation and context, and
- human-in-the-loop intervention and escalation events.

The architecture's role is limited to **producing accurate, attributable telemetry**. It does not compute CSC scores, assign star levels, or interpret assurance outcomes.

Where telemetry is incomplete, delayed, or unavailable, CSC computation shall be constrained, suspended, or withheld, consistent with Version A non-compensatory scoring principles.

6.2 Assurance Boundaries and Interpretation

CSC integration preserves a strict separation between **evidence production** and **assurance interpretation**.

Accordingly:

- CSC scores are computed by governance mechanisms external to the architecture.

- CSC star levels represent **current risk posture and governance maturity**, not quality, performance, or suitability.
- CSC outputs shall be interpreted only within their declared scope, sector profile, and validity context.

No CSC output shall be cached, reused, or displayed without:

- a timestamp,
- a scope declaration, and
- an indication of evidence freshness.

Architecture-specific behavior shall not modify CSC semantics. A change in execution behavior affects CSC only through changes in observed telemetry, not through architectural designation or status. CSC shall therefore be interpreted strictly as a transient governance signal, not as a certification, endorsement, or standing designation.

6.3 Revocation and Degradation Semantics

CSC integration in Version B explicitly supports **assurance degradation and revocation** as first-class governance outcomes.

CSC degradation or revocation may occur when:

- governance controls fail to enforce policy at defined execution boundaries,
- telemetry confidence degrades or becomes incomplete,
- material drift is detected between simulated and observed behavior, or
- human-centric oversight requirements are not met.

Revocation is a **protective control**, not a punitive action. It signals that trust claims are no longer supported by evidence and that corrective action or re-evaluation is required prior to reinstatement.

Re-issuance of CSC assurance shall require renewed evidence generation and evaluation. Administrative reinstatement without supporting telemetry shall not be permitted.

6.4 Evidence Packaging and Proportional Disclosure (Informative)

While CSC assurance is computed as a continuous signal, Version B recognizes that regulatory and audit interactions frequently require **evidence packaging** rather than raw telemetry access.

For this purpose, CSC-relevant signals may be aggregated into governed evidence bundles representing increasing sensitivity and disclosure depth. These bundles are intended to support proportional oversight by enabling:

• confirmation of evidence existence and integrity,
• controlled, time-bound access aligned to authority level, and
• selective disclosure based on necessity rather than convenience.

Importantly, evidence packaging does not alter CSC semantics, scoring logic, or authority. It is an assurance-delivery construct that operates downstream of CSC computation and remains subordinate to the governance framework defined in Version A.

Normative Summary (Chapter 6)

- CSC assurance shall be derived exclusively from telemetry evidence.
- Architecture supplies evidence; governance computes and interprets assurance.
- CSC star levels represent risk posture, not approval or performance.
- Assurance shall be scope-bound, time-bound, and revocable.
- Degradation and revocation are expected governance outcomes when evidence degrades.

---

7. Security, Trust, and Responsibility Boundaries

This section defines the **security, trust, and responsibility boundaries** applicable to the architecture-specific extension introduced in Version B. It clarifies what aspects of system behavior are governed, what aspects are explicitly excluded, and how responsibility is preserved across governance, execution, and assurance layers.

These boundaries are essential to prevent misinterpretation of Version B as a security specification, operational manual, or certification regime.

7.1 What Is Governed in Version B

Version B governs **how governance controls are instantiated at execution boundaries**, not how AI systems are internally designed, optimized, or deployed.

Specifically, Version B governs:

- **Governance execution boundaries**, including where and how policies are enforced at runtime via Access PoD;
- **Telemetry generation and attribution**, ensuring governance-relevant behavior is observable, timestamped, and attributable;
- **Provenance and lineage semantics**, via the UGCCMT model, sufficient to support audit, traceability, and lifecycle reasoning;
- **Controlled execution interfaces**, including API-level safeguards against ungoverned data ingestion or artifact modification; and
- **Assurance interpretation boundaries**, ensuring CSC outputs remain evidence-derived, scope-bound, time-bound, and revocable.

Governance in Version B is concerned with **control effectiveness and evidence integrity**, not internal learning correctness, model quality, or system performance.

7.2 What Is Explicitly Out of Scope

To preserve clarity, safety, and standards integrity, the following are explicitly out of scope for Version B:

- cryptographic algorithms, parameters, key management, or signature schemes;
- validation node internals, trust models, or quorum assumptions;
- security thresholds, drift metrics, re-derivation triggers, or incident response playbooks;
- training algorithms, optimization strategies, or model-internal mechanics;

- deployment, onboarding, or operational procedures for Q-Pathformer;
- performance benchmarking, comparative evaluation, or claims of superiority; and
- regulatory certification, legal compliance determination, or enforcement authority.

The exclusion of these elements is intentional. Their inclusion would blur governance intent with implementation detail and increase security risk without improving assurance clarity.

Certain implementation details related to **governed evidence packaging, controlled disclosure, and third-party confirmation** are intentionally excluded from Version B and may be subject to restricted specification or controlled review. Eligibility, conditions, and cross-validation responsibilities for such disclosures are defined in **Annex A — Access to Non-Public Disclosures**.

7.3 Responsibility Allocation and Non-Transferability

Responsibility within this framework is **explicit, traceable, and non-transferable**.

Accordingly:

- **Governance responsibility** remains with the organization operating Access PoD and asserting conformance to Version A;
- **Execution responsibility** remains with the system operator invoking controlled APIs;
- **Assurance interpretation responsibility** remains with governance and oversight bodies consuming CSC outputs; and
- **Architecture responsibility** is limited to exposing execution boundaries and producing accurate telemetry.

Delegation to external models, tools, or services does **not** transfer responsibility. Where third-party components are used, the integrating organization retains full accountability for governance outcomes within the declared scope.

No architecture, model, or API endpoint may independently assert trust, compliance, or certification.

7.4 Relationship to Restricted Specifications

Certain security-critical mechanisms that support the governance model described in this document are intentionally governed by **restricted specifications**.

These include, but are not limited to:

- cryptographic realization of provenance and audit artifacts;
- validation and sealing mechanisms;
- key lifecycle and revocation controls; and
- jurisdiction-specific security adaptations.

Version B references these mechanisms only at a conceptual level to maintain completeness without disclosing sensitive implementation detail. Access to restricted specifications does not affect the validity or applicability of this public standards-exploration document.

7.5 Relationship to Version C and Operational Documentation

Version B is **not** an onboarding guide, operational manual, or implementation specification.

A subsequent **Version C / Volume 2** document is planned to address:

- Q-Pathformer onboarding workflows;
- operational mechanisms and execution configuration; and
- architecture-specific implementation detail.

Version C will remain subordinate to the governance framework defined in Version A and instantiated in Version B. It will not redefine governance authority, assurance semantics, or CSC scoring logic.

Normative Summary (Chapter 7)

- Version B governs execution-level governance instantiation, not system internals.
- Cryptographic and security realizations are intentionally out of scope.
- Responsibility is explicit and non-transferable.
- Delegation does not dilute accountability.
- Controlled disclosure is governed by Annex A.
- Operational onboarding is reserved for Version C.

---

8. Forward-Looking Considerations (Non-Normative)

This chapter outlines forward-looking considerations related to interoperability, cryptographic evolution, and lifecycle transition beyond Version B. It is **informative only** and does not introduce new requirements, controls, or governance obligations.

The purpose of this chapter is to situate the Version B extension within a longer-term governance trajectory while maintaining strict separation between **validated practice** and **future exploration**.

8.1 Interoperability and Ecosystem Evolution

As AI systems increasingly operate within multi-organizational, multi-model, and cross-jurisdictional ecosystems, governance mechanisms must support interoperability without imposing uniform architecture or centralized control.

Within this framework:

- Governance signals (e.g., telemetry summaries, CSC outputs, sector profiles) are intended to be **machine-readable and portable**.
- Interoperability refers to the **interpretability of governance evidence**, not the uniformity of system design.
- Multiple architectures and model providers may coexist under shared governance expectations, provided that evidence generation and scope declaration remain explicit.

The Access PoD framework is designed to support this evolution by functioning as a **governance overlay**, rather than a tightly coupled execution platform.

8.2 Future Cryptographic and Privacy Enhancements

Version B intentionally excludes cryptographic realization, privacy-preserving computation, and advanced security mechanisms from public specification.

Future work may explore, subject to appropriate review and standards development:

- enhanced cryptographic provenance anchoring,
- privacy-preserving verification techniques (e.g., selective disclosure),
- quantum-resistant assurance mechanisms, and
- scalable validation infrastructure for high-volume governance telemetry.

Any such enhancements would be governed by **separate specifications** and introduced only after sufficient maturity, review, and validation. Their inclusion would not alter the governance semantics or assurance principles defined in Version A or instantiated in Version B.

Cryptographic techniques, including encryption, may be employed to support confidentiality, integrity, or selective disclosure of governance evidence; however, their use is **implementation-specific** and **not a prerequisite** for conformance, assurance validity, or interoperability under the Version A framework.

8.3 Transition Considerations for Version C

Version B is intentionally not an onboarding or operational guide.

A subsequent **Version C / Volume 2 document** is planned to address architecture-specific onboarding, operational workflows, and implementation mechanisms related to the Q-Pathformer system.

That document will:

- remain subordinate to the governance framework defined in Version A,
- preserve the authority and semantics of CSC assurance, and
- provide practical guidance for deployment without redefining trust or compliance criteria.

The separation between Version B and Version C ensures that governance validation, standards exploration, and operational enablement remain **independently reviewable**.

8.4 Version D — Governance Stress-Testing and Ethical Evaluation (Non-Normative)

A subsequent Version D is envisaged to introduce structured governance stress-testing, including ethical and adversarial evaluation of new model configurations, delegation pathways, and policy boundary conditions. These activities are intended to surface governance failure modes—such as policy bypass, telemetry suppression, or erosion of human-in-the-loop controls—that may not manifest under nominal operation.

Version D activities shall operate under explicit standards alignment, published terms and conditions, and defined ethical testing rules of engagement. Findings are treated as governance evidence, not exploit disclosures, and are used to inform policy refinement, assurance constraints, and lifecycle decisions rather than deployment approval.

Informative Summary (Chapter 8)

- Interoperability concerns governance evidence, not architecture uniformity.
- Cryptographic and privacy enhancements are future considerations, not current requirements.
- Operational onboarding is reserved for Version C.
- Governance semantics defined in Version A remain stable across future extensions.

---

9. Conclusion

This document has presented an **architecture-specific extension** to the Volume 2 governance framework defined in Version A, using the Q-Pathformer system as a reference implementation.

The extension demonstrates that:

- runtime governance controls can be instantiated at execution depth without embedding governance authority in system architecture,
- continuous, telemetry-derived assurance is feasible for adaptive AI systems, and
- trust can be engineered, observed, degraded, and revoked based on evidence rather than declaration.

Version B preserves the architecture-agnostic integrity of the original framework while providing concrete grounding for reviewers, policymakers, and system designers seeking to evaluate governance feasibility beyond abstraction.

9.1 Engineering Trust Before Impact

Trust in AI systems shall not be inferred from documentation, intent, or reputation. It must be **engineered before impact occurs** and sustained through continuous observation and control.

By integrating:

- simulation-first risk discovery (SAPP),
- runtime policy enforcement (Access PoD), and
- evidence-based assurance (CSC),

this framework enables governance to operate proactively rather than reactively, reducing systemic risk while supporting responsible innovation.

9.2 Standards, Conformance, and Ongoing Evolution

This document does not claim to define a final or exclusive standard.

Instead, it provides a **reference architecture extension** through which governance practices may be tested, compared, and refined. Standards leadership is expected to emerge from **validated, repeatable practice**, not assertion.

Conformance to this framework:

- shall be evidence-based,
- shall be continuously evaluated,
- shall be scope-bound and revocable, and
- shall remain independent of any single architecture or provider.

As AI systems and regulatory expectations evolve, the principles articulated in this document are intended to support convergence toward interoperable, lifecycle-aware governance infrastructure.

Final Normative Statement

- Trust shall be engineered, observed, and maintained continuously.
- Governance shall be executed at runtime, not asserted post hoc.
- Assurance shall be evidence-based and revocable.
- Architecture shall support governance without defining it.
- Standards shall evolve from validated practice, not declaration.

Conformance and Interoperability Clause (Version B)

Conformance to the governance framework described in *A WEB of Sharing Trust–Responsibility Toward a Trustless Future* is defined **exclusively by Volume 2, Version A**.

Version B does **not** introduce new conformance criteria, compliance requirements, certification conditions, or assessment thresholds. The architecture-specific content presented in this edition is provided solely to illustrate how the Version A governance framework **may be instantiated at execution depth** within a concrete system, without altering governance authority, assurance semantics, or scoring logic.

Interoperability, in the context of Version B, refers to the ability of **governance evidence, telemetry semantics, and assurance signals** to be interpreted consistently across heterogeneous systems, organizations, and regulatory environments. It does **not** imply interoperability of model architectures, training methods, cryptographic techniques, or execution substrates.

The adoption, modification, or omission of any architecture-specific mechanism described in Version B—including execution APIs, provenance abstractions, or optional security techniques—shall not be interpreted as affecting conformance, certification status, or regulatory standing under the Version A framework.

Version B is therefore **illustrative and non-binding with respect to conformance**, and shall be read as a standards-exploration reference rather than a normative compliance specification.

Annex A — Access to Non-Public Disclosures

(Informative — Non-Normative)

A.1 Purpose and Scope

This annex defines the **eligibility criteria, access conditions, and review responsibilities** associated with non-public disclosures referenced in *A WEB of Sharing Trust–Responsibility Toward a Trustless Future, Volume 2 — Version B*.

The purpose of this annex is to ensure that:

- security-sensitive mechanisms are reviewed **accurately and responsibly**;
- disclosure practices support **standards convergence and regulatory oversight**;
- implementation safety is preserved during iterative validation and refinement.

This annex governs **review access only**. It does not define system behavior, compliance requirements, or operational obligations.

A.2 Principles of Controlled Disclosure

Non-public disclosure within this framework is governed by the following principles:

a) **Safety before completeness**
Certain mechanisms are unsafe or misleading if disclosed partially or without full context.

b) **Evidence over assertion**
Disclosure evolves based on validated operation and ecosystem maturity, not design intent alone.

c) **Separation of governance and implementation**
Public materials define governance intent and assurance semantics; non-public materials clarify implementation realization for review accuracy.

d) **No privileged trust**
Access to non-public materials does not confer endorsement, authority, or compliance status.

A.3 Eligible Parties for Non-Public Access

Access to non-public disclosures is **role-based**, not organization-based.

A.3.1 Government and Regulatory Authorities

Government agencies and regulatory bodies may be granted access where required for:

- regulatory evaluation or policy development;
- oversight feasibility analysis;
- post-deployment governance assessment.

Such access supports informed oversight and proportional enforcement design.
No approval, endorsement, or certification is implied.

A.3.2 Standards Bodies and Governance Institutions

Standards organizations, technical committees, and governance councils may access non-public materials for:

- evaluation of candidate mechanisms for guidance, technical reports, or future standards;
- assessment of interoperability and scope discipline;
- identification of safety and misuse considerations.

Access is intended to support **standards refinement**, not premature standardization.

A.3.3 Private Enterprises and System Operators

Private enterprises and infrastructure operators may access non-public materials when:

- evaluating adoption feasibility under operational constraints;
- conducting internal risk, security, or compliance assessments;
- participating in controlled validation or pilot activities.

Access is limited to governance-relevant mechanisms and does not extend to competitive model internals or optimization strategies.

A.3.4 Academic and Independent Reviewers

Academic or independent reviewers may be granted limited access when contributing to:

- governance, safety, or assurance research;
- structured peer review activities;
- standards-adjacent validation programs.

Access scope is restricted to conceptual correctness, misuse resistance, and interpretability.

A.4 Conditions for Granting Non-Public Access

Non-public disclosure is subject to the following conditions:

a) **Declared review purpose**
The requesting party shall state the specific review objective.

b) **Scope limitation**
Access shall be limited to materials necessary for the stated purpose.

c) **Non-deployment commitment**
Non-public materials shall not be operationalized or represented as production-ready.

d) **Context preservation**
Materials shall be reviewed with full contextual explanation to avoid misinterpretation.

A.5 Infrastructure Layers and Mutual Cross-Validation

The framework enables cross-validation through **intentional separation of layers**.

A.5.1 Governance Layer (Access PoD)

- Defines governance proposals and constraints.
- Enforces policy at execution boundaries.
- Emits governance telemetry.
- Does not execute learning or inference.

A.5.2 Execution Layer (Architecture-Specific)

- Executes only approved actions.

- Generates provenance and lifecycle signals.
- Cannot approve governance decisions.
- Does not compute assurance outcomes.

A.5.3 Assurance Layer (CSC)

- Interprets telemetry and simulation outputs.
- Produces continuous, revocable assurance signals.
- Cannot influence execution or governance proposals.

A.5.4 Evidence Delivery Layer (Optional)

- Packages evidence proportionally.
- Governs disclosure by authority, time, and necessity.
- Enables third-party confirmation without content exposure.

Each layer is independently observable and **mutually constraining**, enabling validation of intent, direction, and operational validity.

A.6 Cross-Validation Responsibilities

Access to non-public disclosures carries an expectation of **active validation**.

A.6.1 Validation of Intent

Reviewers may assess alignment between disclosed mechanisms and stated governance objectives.

A.6.2 Validation of Direction

Reviewers may evaluate whether proposed evolution paths are technically and institutionally viable.

A.6.3 Validation of Validity

Reviewers may challenge assumptions regarding threat models, misuse resistance, and failure modes.

Feedback is expected to inform iterative refinement cycles.

A.7 Disclosure Evolution and Validation Cadence

This framework assumes **continuous advancement**, not static specification.

- Enhancements are proposed and reviewed on a recurring cadence (e.g., quarterly).
- Disclosure status may evolve based on demonstrated safety, maturity, and validation outcomes.
- Mechanisms may transition from non-public to public disclosure only after sustained validation.

A.8 Distribution Ledgers

The use of a distribution or anchoring ledger is:

- optional,
- non-normative, and
- not required for governance or assurance validity.

Ledger adoption, if any, is an implementation choice subject to separate review and validation.

A.9 No Transfer of Authority or Trust

Non-public access does not grant:

- governance authority,
- endorsement rights,
- compliance status, or
- preferential treatment.

Trust within this framework remains evidence-derived, revocable, and non-transferable.

A.10 Relationship to Main Document

This annex supplements, but does not modify, the normative content of the main document.

References to controlled disclosure within Section 7 of Version B should be interpreted in conjunction with this annex.

Informative Summary

Controlled disclosure supports responsible governance evolution by enabling accurate review of sensitive mechanisms while preserving safety, interoperability, and public trust.

Annex B — Informative References and Supporting Artifacts

(Non-Normative)

Purpose of this Annex

This annex provides an index of **supporting artifacts referenced by Version B**, intended to assist reviewers, regulators, standards bodies, and system operators in understanding how the governance framework may be evaluated, tested, and interrogated in practice.

Inclusion of an artifact in this annex:

- does **not** imply certification, approval, or compliance status;
- does **not** require adoption of any specific implementation; and
- does **not** alter the normative scope of this document.

Certain artifacts are descriptive or illustrative. Others are suitable for structured review under controlled conditions.

Notation:
Artifacts are annotated for reviewer orientation only:
[P] indicates public, human-readable materials; **[S]** indicates schemas or structural definitions; **[R]** indicates review-grade reference artifacts intended for structured evaluation or traceability discussion.
These annotations do not imply access control, confidentiality classification, or normative status.

B.1 Governance Evidence Manifests

- **apod.evidence_manifest.json [R][S]**
  Reference manifest describing how governance evidence artifacts may be grouped, identified, and contextualized for review.
  This file is descriptive only and does not constitute an operational ledger or disclosure mechanism.
- **apod.evidence_manifest.schema.json [S]**
  Schema defining the structure of the evidence manifest for consistency checking and tooling interoperability.

B.2 Compliance Star Certificate (CSC) Reference Artifacts

- **csc_snapshot.json [P][S]**
  Example snapshot illustrating the structure and scope metadata of a CSC assurance output.
- **csc_snapshot.schema.json [S]**
  Schema defining CSC snapshot fields for interpretability and validation tooling.
- **csc_bundle_reference.json [P][S]**
  Descriptive reference illustrating how CSC signals may be grouped into proportional evidence bundles.
  This artifact does not describe cryptographic enforcement, decryption logic, or disclosure authorization.

B.3 EU AI Act Annex VII Alignment Artifacts

- **apod.self_assessment.eu_annex_vii.json [P]**
  High-level self-assessment mapping the governance framework to Annex VII concepts.
- **apod.self_assessment.eu_annex_vii.expanded.non-native.json [R]**
  Expanded internal mapping used for review and traceability discussions.
  This artifact is illustrative and does not constitute a legal determination.
- **apod.self_assessment.eu_annex_vii.schema.json [S]**
  Schema supporting structured comparison and consistency checking.
- **EU AI Act Annex VII-Aligned Evidence Checklist (PDF) [P]**
  Human-readable checklist supporting regulatory dialogue and gap analysis.

B.4 Operator and Oversight Materials

- **Operator Evidence Checklist – Pre-Regulatory Engagement (PDF) [P]**
  Checklist intended to support internal readiness and structured engagement with regulators.
- **Operator's Manual – Access PoD (PDF) [R]**
  High-level operational reference for governance execution boundaries.
  Does not disclose cryptographic enforcement or security-sensitive mechanisms.
- **Regulator Q&A Playbook (PDF) [P]**
  Anticipated regulatory questions with approved explanatory responses.
- **Board Briefing — Runtime AI Governance & Risk Oversight (PDF) [P]**
  Executive-level briefing focusing on risk, accountability, and governance posture.

B.5 Notes on Artifact Use and Interpretation

Artifacts listed in this annex are provided to support understanding, evaluation, and informed discussion.

They do not represent mandatory components, final specifications, or production-ready implementations.

Where artifacts are designated as review-grade, access and use may be subject to controlled disclosure conditions described elsewhere in this document.

Clarifying Inserts (Informative — Non-Normative)

Public Review Artifact Pack (Version B)

The public review artifact packs accompanying this document provide supporting, non-normative materials intended to assist reviewers, regulators, standards bodies, and system operators in evaluating the governance framework in practice.

Two artifact distributions are currently maintained:

1. Access PoD Artifact Pack — v1b (Public Review Draft) released early Jan 2026.
This package contains the Version B–specific illustrative artifacts referenced throughout this document, including governance evidence manifests, Compliance Star Certificate (CSC) reference structures, EU AI Act Annex VII–aligned materials, and operator- and regulator-facing review aids.

Canonical release (current):
https://github.com/Access-PoD/access-pod-artifacts/releases/tag/v1b

2. Access PoD Composite Workspace Pack — v1.0-workspace
In addition to the standalone artifact pack, an early composite workspace package is provided to support execution-level review, reproducibility validation, and governance inspection.

This composite package includes:

- the original **v1a** public review artifacts,
- the **v1b** public review artifacts,
- both sets unpacked into a deterministic, inspectable workspace layout, and
- a governance harness supporting **fetch, unpack, and verify** operations.

The harness is designed to:

- fetch tagged Access PoD artifact releases,
- verify integrity and lineage prior to use,
- unpack artifacts into a controlled workspace, and
- support future tagged releases without modifying existing artifacts.

The composite package is intended solely for inspection, testing, and standards review.
It does **not** imply certification, approval, operational readiness, or deployment endorsement.

Canonical release (composite workspace):
https://github.com/Access-PoD/access-pod-artifacts/releases/tag/v1.1-workspace

Both packages are maintained concurrently to preserve lineage clarity and to allow reviewers to choose between:

- artifact-level inspection (standalone packs), and
- workspace-level execution and verification review (composite pack).

Availability of either package does not imply that any specific artifact, harness, or workflow is required for conformance. All materials remain illustrative and non-normative.

References

(Informative — Non-Normative)

The following documents are referenced for context, alignment, and further information. For dated references, only the edition cited applies. For undated references, the latest edition applies.

[1] ISO/IEC 42001
Information technology — Artificial intelligence — Management system.
International Organization for Standardization (ISO) / International Electrotechnical Commission (IEC).

[2] ISO/IEC 27001
Information security, cybersecurity and privacy protection — Information security management systems — Requirements.
International Organization for Standardization (ISO) / International Electrotechnical Commission (IEC).

[3] Regulation (EU) 2024/… (AI Act)
Regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence.
European Union.

[4] A WEB of Sharing Trust–Responsibility Toward a Trustless Future — Volume 2, Version A
Anthony Siljeg.
Status: Public Review Draft (Standards Exploration).

[5] Policy of Developments / Access PoD Initiative
Project website and public materials.
https://policyofdevelopments.org/

Inclusion of these references does not imply endorsement, certification, or conformance.

Cover Note — Regulator Briefing (Informative)

This Regulator Briefing is included as an **informative, non-normative supplement** to *A WEB of Sharing Trust–Responsibility Toward a Trustless Future, Volume 2 — Version B*.

The purpose of this briefing is to support **regulatory understanding and review** by summarizing the governance intent, assurance mechanics, and oversight implications of the framework in a form tailored to public-sector and standards audiences.

This briefing:

- does **not** introduce new requirements, controls, or obligations;
- does **not** modify or extend the normative content of the main document or appendices;
- does **not** constitute regulatory guidance, certification criteria, or legal advice; and
- does **not** assume adoption, endorsement, or approval by any regulatory body.

The main body of this document defines the governance framework and its architecture-specific instantiation using standards-aligned language. Appendices provide methodological detail where appropriate. The Regulator Briefing is provided **in addition to**, not in place of, those sections.

How Regulators Are Encouraged to Read This Briefing

Regulators are encouraged to treat this briefing as:

- a **contextual summary** of how the framework addresses known gaps in static, documentation-based AI oversight;
- a **translation layer** between standards-style governance language and regulatory oversight concerns; and
- a **non-binding reference** for evaluating whether evidence-based, runtime governance approaches may complement existing regulatory tools.

The briefing is intentionally high-level. It emphasizes **observable system behavior, continuous assurance, and revocation semantics**, rather than technical implementation detail. Where detailed mechanisms (e.g., cryptographic controls or operational procedures) are referenced, they are governed by restricted specifications or future documentation and are not required for regulatory interpretation of this document.

Scope and Authority

Nothing in this briefing alters the scope, authority, or applicability of the main document. Regulatory authority remains with the relevant oversight bodies, and responsibility for governance outcomes remains with system operators asserting conformance to the framework.

The inclusion of this briefing reflects the author's intent to **facilitate transparent, informed regulatory dialogue** as AI governance practices evolve toward runtime, evidence-based models.

How This Framework Differs from Certification Schemes

(Informative — Clarifying Note for Regulators)

This document and its associated mechanisms are **not a certification scheme** and should not be interpreted as such.

The framework described in this volume differs from traditional certification models in several fundamental ways.

1. Certification vs. Continuous Assurance

Traditional certification schemes:

- assess systems at a point in time,
- rely on documentation, attestations, and audits, and
- issue a status that remains valid until expiry or revocation through external action.

By contrast, this framework:

- derives assurance **continuously** from observed system behavior,
- treats trust as **time-bound and evidence-dependent**, and
- supports automatic degradation or revocation when evidence no longer supports prior trust claims.

If telemetry does not exist, **assurance does not exist**.

---

2. Status Labels vs. Evidence-Derived Signals

Certification schemes typically issue:

- binary or tiered status labels (certified / not certified),
- often reused across contexts and deployments.

The Compliance Star Certificate (CSC):

- is a **computed assurance signal**, not a designation,

- is explicitly **scope-bound, time-stamped, and contextual**, and
- cannot be reused, transferred, or cached without fresh evidence.

CSC star levels represent **current governance risk posture**, not approval, quality, or fitness for purpose.

---

3. Approval Authority vs. Governance Infrastructure

Certification schemes typically:

- assert approval authority, and
- function as gatekeepers for market access or compliance claims.

This framework:

- does **not** approve systems,
- does **not** confer regulatory status, and
- does **not** replace sector-specific compliance regimes.

It provides **governance infrastructure** that regulators may observe, interrogate, or integrate with existing oversight mechanisms.

---

4. Static Compliance vs. Runtime Oversight

Certification schemes are usually:

- document-centric,
- periodic, and
- retrospective.

This framework is:

- execution-centric,
- runtime-observable, and
- forward-looking.

It enables regulators to evaluate **how systems behave under real conditions**, including drift, misuse, and control degradation - rather than relying on declared intent.

Regulatory Interpretation

Regulators should therefore interpret this framework as:

- a **complement to**, not a replacement for, certification and compliance schemes;
- an **evidence-generation and assurance substrate**, not an approval mechanism; and
- a tool for **continuous oversight**, not one-time authorization.

Informative Summary

- This framework is not a certification scheme.
- It does not issue approvals or compliance status.
- It produces continuous, revocable assurance signals from evidence.
- It is designed to support runtime oversight in adaptive AI systems.

Regulator Briefing (Informative — Non-Normative)

(Informative — Regulator-Facing Summary)

Purpose of This Briefing

This briefing is provided to assist **regulators, standards bodies, and public-sector oversight authorities** in understanding the governance intent, assurance mechanics, and regulatory relevance of *A WEB of Sharing Trust–Responsibility Toward a Trustless Future, Volume 2 — Version B*.

It is **not normative**, does not introduce new requirements, and does not constitute regulatory guidance or legal advice. Its sole purpose is to support informed regulatory review.

---

1. Problem Statement: Limits of Existing AI Oversight Models

Current AI oversight mechanisms rely heavily on:

- documentation and attestations,
- point-in-time audits,
- static risk registers, and
- post-incident investigations.

These mechanisms are poorly suited to modern AI systems that are:

- continuously retrained or updated,
- composed of multiple models and external services,
- deployed across jurisdictions and organizational boundaries, and
- capable of emergent behavior over time.

As a result, regulators face a **visibility gap** between what systems are declared to do and how they actually behave during operation.

---

2. Core Regulatory Shift Introduced by This Framework

This framework proposes a shift from **compliance as declaration** to **assurance as evidence**.

Rather than asking:

“Does the system claim to meet requirements?”

the framework enables regulators to ask:

“What does the system demonstrably do, under real conditions, over time?”

This is achieved through three integrated mechanisms:

1. **Simulation AI-Powered Policies (SAPP)** — pre-deployment and early-deployment risk discovery,
2. **Access Policy-on-Demand (PoD)** — runtime enforcement and telemetry generation, and
3. **Compliance Star Certificate (CSC)** — continuous, revocable assurance derived from evidence.

---

3. What Regulators Can Observe (and What They Do Not Need to Trust)

Under this model, regulators are not asked to trust:

- internal model architecture,
- proprietary training techniques,
- cryptographic implementation detail, or
- vendor claims of “responsible AI.”

Instead, regulators can observe:

- **runtime governance telemetry**, including policy enforcement outcomes,
- **simulation evidence** showing how systems behave under stress or misuse,
- **human-in-the-loop intervention records**, and
- **assurance degradation or revocation events** when controls fail.

If telemetry does not exist, **assurance does not exist**.

---

4. Compliance Star Certificate (CSC): What It Is and What It Is Not

What CSC Is

- A **dynamic assurance signal**, not a certification.
- Computed continuously from runtime and simulation telemetry.
- Explicitly scope-bound, time-bound, and revocable.
- Aligned with ISO/IEC risk-management principles.

CSC star levels represent **current governance risk posture**, not system quality, performance, or approval.

What CSC Is Not

- Not a regulatory certification.
- Not a compliance label.
- Not a one-time approval.
- Not transferable without scope and evidence review.

This avoids the common regulatory pitfall of **certification-as-status**.

---

5. Architecture-Specific Reference (Version B Context)

Version B introduces a **reference implementation** using the Q-Pathformer system to demonstrate feasibility at execution depth.

This reference implementation is used solely to show:

- how governance attaches to execution boundaries,
- how telemetry is generated from real system behavior, and
- how catastrophic data or model overwrite can be prevented through API-level governance.

Importantly:

- Governance authority **does not reside in the architecture**.
- Regulators are not required to endorse or adopt the reference system.
- The governance framework remains architecture-agnostic.

---

6. Regulatory Advantages of This Approach

For regulators, this model provides:

- **Continuous oversight**, not episodic audits.
- **Early warning signals** through simulation and telemetry.
- **Evidence-based enforcement**, not documentation disputes.
- **Clear revocation semantics** when trust degrades.
- **Interoperability**, allowing oversight across vendors and sectors.

This approach is compatible with:

- EU AI Act lifecycle risk management,
- ISO/IEC 42001 AI management systems,
- sector-specific safety and accountability regimes.

---

7. What This Framework Does Not Do

This framework does **not**:

- replace sector-specific regulation,
- define legal compliance thresholds,
- mandate specific technical architectures,
- expose sensitive security or cryptographic mechanisms, or
- shift responsibility away from system operators.

It provides **governance infrastructure**, not regulatory authority.

---

8. Key Takeaway for Regulators

This framework enables regulators to move from:

"Trust us — we comply."

to:

"Here is what the system actually did, when, under which controls, and with what outcome."

Trust becomes **observable**, **measurable**, and **revocable**.

Regulator Summary (Non-Normative)

- AI governance must operate at runtime, not only at design time.
- Assurance must be evidence-based, not declarative.
- Certification must be revocable when evidence degrades.
- Oversight must scale with system adaptation and deployment.

This document provides a reference pathway toward that outcome.